Groups and Lie Algebras: Exercise Sheet 10

1 The Killing form

- Compute the Killing form of $\mathfrak{sl}(2)$ in the Cartan basis H, E, F.
- Show that the Killing form satisfies $\kappa([X,Y],Z) = \kappa(X,[Y,Z])$.
- Show that the Killing form is invariant under any automorphism of the Lie algebra $\mathfrak{g}$.
- Let $\{T^a\}_{a=1,\dots,n}$ be a basis of an n-dimensional Lie algebra $\mathfrak{g}$ with structure constants $f^{ab}{}_c$. Show that the Killing form can be expressed as

$$\kappa(T^a, T^b) = \sum_{c,e=1}^{n} f^{bc}{}_e \, f^{ae}{}_c. \tag{1}$$

- Use this to prove formula (*) from the lecture (Lecture notes page 7).

2 The root diagram and Weyl group of G_2
Starting from the G_2 Dynkin diagram draw the simple roots and compute the Weyl group. Use this to sketch out the entire root system of G_2.

3 The YM action functional
Let G be a simple Lie group with (simple) Lie algebra $\mathfrak{g}$. The fields of G-Yang–Mills theory on $\mathbb{R}^4$ are 1-forms with values in $\mathfrak{g}$. In case you don't know what a 1-form is: They are just described by 4 functions $A_i\colon \mathbb{R}^4 \to \mathfrak{g}$ for $i = 1, \dots, 4$ representing the 1-form $A = \sum_i A_i \, dx^i$. The field strength 2-form $F = dA + A \wedge A$ is explicitly given by the 16 (they are antisymmetric in i and j and hence there are only 6 independent functions) functions $F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j]$ which are maps from $\mathbb{R}^4 \to \mathfrak{g}$. The action functional of Yang–Mills theory is

$$S_{\mathrm{YM}}[A] = -\frac{1}{4e^2} \sum_{i,j} \int_{\mathbb{R}^4} \kappa\big(F_{ij},\, F_{ij}\big)\, d^4x, \tag{2}$$

where e is the coupling constant. A gauge transformation is a smooth map $g\colon \mathbb{R}^4 \to G$ and acts on the fields by:

$$A_i \;\longmapsto\; A_i^g = g\, A_i\, g^{-1} - (\partial_i g)\, g^{-1}, \tag{3}$$

Writing gauge transformations like this is standard in physics, but needs some explanation: the first term is the adjoint action of G on $\mathfrak{g}$. To understand the second term note that $\partial_i g$ is a tangent vector at g. Multiplying with g^{-1} gives a tangent vector at the identity, i.e. an element of $T_e G = \mathfrak{g}$.

a) Make sure that you understand the above definitions and recall the necessary background from the lecture.

b) Work out the case of $G = SU(2)$ explicitly. It is up to you to decide what this means.

c) Compute the transformation of F under gauge transformations.

d) Show that $S_{\mathrm{YM}}[A]$ is gauge invariant, i.e. $S_{\mathrm{YM}}[A] = S_{\mathrm{YM}}[A^g]$.